# THE ART OF ELECTRONICS

PAUL HOROWITZ
WINFIELD HILL

THIRD EDITION

# The Art of Electronics

## Third Edition

At long last, here is the thoroughly revised and updated, and long-anticipated, third edition of the hugely successful *The Art of Electronics*. Widely accepted as the best single authoritative text and reference on electronic circuit design, both analog and digital, the first two editions were translated into eight languages, and sold more than a million copies worldwide. The art of electronics is explained by stressing the methods actually used by circuit designers – a combination of some basic laws, rules of thumb, and a nonmathematical treatment that encourages understanding why and how a circuit works.

Paul Horowitz is a Research Professor of Physics and of Electrical Engineering at Harvard University, where in 1974 he originated the Laboratory Electronics course from which emerged *The Art of Electronics*. In addition to his work in circuit design and electronic instrumentation, his research interests have included observational astrophysics, x-ray and particle microscopy, and optical interferometry. He is one of the pioneers of the search for intelligent life beyond Earth (SETI). He is the author of some 200 scientific articles and reports, has consulted widely for industry and government, and is the designer of numerous scientific and photographic instruments.

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Third Edition

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Winfield Hill ROWLAND INSTITUTE AT HARVARD

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*To Vida and Ava*

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*In Memoriam: Jim Williams, 1948–2011*

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# PREFACE TO THE FIRST EDITION

This volume is intended as an electronic circuit design textbook and reference book; it begins at a level suitable for those with no previous exposure to electronics and carries the reader through to a reasonable degree of proficiency in electronic circuit design. We have used a straightforward approach to the essential ideas of circuit design, coupled with an in-depth selection of topics. We have attempted to combine the pragmatic approach of the practicing physicist with the quantitative approach of the engineer, who wants a thoroughly evaluated circuit design.

This book evolved from a set of notes written to accompany a one-semester course in laboratory electronics at Harvard. That course has a varied enrollment – undergraduates picking up skills for their eventual work in science or industry, graduate students with a field of research clearly in mind, and advanced graduate students and postdoctoral researchers who suddenly find themselves hampered by their inability to "do electronics."

It soon became clear that existing textbooks were inadequate for such a course. Although there are excellent treatments of each electronics specialty, written for the planned sequence of a four-year engineering curriculum or for the practicing engineer, those books that attempt to address the whole field of electronics seem to suffer from excessive detail (the handbook syndrome), from oversimplification (the cookbook syndrome), or from poor balance of material. Much of the favorite pedagogy of beginning textbooks is quite unnecessary and, in fact, is not used by practicing engineers, while useful circuitry and methods of analysis in daily use by circuit designers lie hidden in application notes, engineering journals, and hard-to-get data books. In other words, there is a tendency among textbook writers to represent the theory, rather than the art, of electronics.

We collaborated in writing this book with the specific intention of combining the discipline of a circuit design engineer with the perspective of a practicing experimental physicist and teacher of electronics. Thus, the treatment in this book reflects our philosophy that electronics, as currently practiced, is basically a simple art, a combination of some basic laws, rules of thumb, and a large bag of tricks. For these reasons we have omitted entirely the usual discussions of solid-state physics, the *h*-parameter model of transistors, and complicated network theory, and reduced to a bare minimum the mention of load lines and the *s*-plane. The treatment is largely nonmathematical, with strong encouragement of circuit brainstorming and mental (or, at most, back-of-the-envelope) calculation of circuit values and performance.

In addition to the subjects usually treated in electronics books, we have included the following:

- an easy-to-use transistor model;
- extensive discussion of useful subcircuits, such as current sources and current mirrors;
- single-supply op-amp design;
- easy-to-understand discussions of topics on which practical design information is often difficult to find: op-amp frequency compensation, low-noise circuits, phase-locked loops, and precision linear design;
- simplified design of active filters, with tables and graphs;
- a section on noise, shielding, and grounding;
- a unique graphical method for streamlined low-noise amplifier analysis;
- a chapter on voltage references and regulators, including constant current supplies;
- a discussion of monostable multivibrators and their idiosyncrasies;
- a collection of digital logic pathology, and what to do about it;
- an extensive discussion of interfacing to logic, with emphasis on the new NMOS and PMOS LSI;
- a detailed discussion of A/D and D/A conversion techniques;
- a section on digital noise generation;
- a discussion of minicomputers and interfacing to data buses, with an introduction to assembly language;
- a chapter on microprocessors, with actual design examples and discussion – how to design them into instruments, and how to make them do what you want;
- a chapter on construction techniques: prototyping, printed circuit boards, instrument design;

- a simplified way to evaluate high-speed switching circuits;
- a chapter on scientific measurement and data processing: what you can measure and how accurately, and what to do with the data;
- bandwidth narrowing methods made clear: signal averaging, multichannel scaling, lock-in amplifiers, and pulse-height analysis;
- amusing collections of "bad circuits," and collections of "circuit ideas";
- useful appendixes on how to draw schematic diagrams, IC generic types, *LC* filter design, resistor values, oscilloscopes, mathematics review, and others;
- tables of diodes, transistors, FETs, op-amps, comparators, regulators, voltage references, microprocessors, and other devices, generally listing the characteristics of both the most popular and the best types.

Throughout we have adopted a philosophy of naming names, often comparing the characteristics of competing devices for use in any circuit, and the advantages of alternative circuit configurations. Example circuits are drawn with real device types, not black boxes. The overall intent is to bring the reader to the point of understanding clearly the choices one makes in designing a circuit – how to choose circuit configurations, device types, and parts values. The use of largely nonmathematical circuit design techniques does not result in circuits that cut corners or compromise performance or reliability. On the contrary, such techniques enhance one's understanding of the real choices and compromises faced in engineering a circuit and represent the best approach to good circuit design.

This book can be used for a full-year electronic circuit design course at the college level, with only a minimum mathematical prerequisite; namely, some acquaintance with trigonometric and exponential functions, and preferably a bit of differential calculus. (A short review of complex numbers and derivatives is included as an appendix.) If the less essential sections are omitted, it can serve as the text for a one-semester course (as it does at Harvard).

A separately available laboratory manual, *Laboratory Manual for the Art of Electronics* (Horowitz and Robinson, 1981), contains twenty-three lab exercises, together with reading and problem assignments keyed to the text.

To assist the reader in navigation, we have designated with open boxes in the margin those sections within each chapter that we feel can be safely passed over in an abbreviated reading. For a one-semester course it would probably be wise to omit, in addition, the materials of Chapter 5 (first half), 7, 12, 13, 14, and possibly 15, as explained in the introductory paragraphs of those chapters.

We would like to thank our colleagues for their thoughtful comments and assistance in the preparation of the manuscript, particularly Mike Aronson, Howard Berg, Dennis Crouse, Carol Davis, David Griesinger, John Hagen, Tom Hayes, Peter Horowitz, Bob Kline, Costas Papaliolios, Jay Sage, and Bill Vetterling. We are indebted to Eric Hieber and Jim Mobley, and to Rhona Johnson and Ken Werner of Cambridge University Press, for their imaginative and highly professional work.

*Paul Horowitz*
*Winfield Hill*
*April 1980*

# PREFACE TO THE SECOND EDITION

Electronics, perhaps more than any other field of technology, has enjoyed an explosive development in the last four decades. Thus it was with some trepidation that we attempted, in 1980, to bring out a definitive volume teaching the art of the subject. By "art" we meant the kind of mastery that comes from an intimate familiarity with real circuits, actual devices, and the like, rather than the more abstract approach often favored in textbooks on electronics. Of course, in a rapidly evolving field, such a nuts-and-bolts approach has its hazards – most notably a frighteningly quick obsolescence.

The pace of electronics technology did not disappoint us! Hardly was the ink dry on the first edition before we felt foolish reading our words about "the classic [2Kbyte] 2716 EPROM. . . with a price tag of about $25." They're so classic you can't even get them anymore, having been replaced by EPROMs 64 times as large, and costing less than half the price! Thus a major element of this revision responds to improved devices and methods – completely rewritten chapters on microcomputers and microprocessors (using the IBM PC and the 68008) and substantially revised chapters on digital electronics (including PLDs, and the new HC and AC logic families), on op-amps and precision design (reflecting the availability of excellent FET-input op-amps), and on construction techniques (including CAD/CAM). Every table has been revised, some substantially; for example, in Table 4.1 (operational amplifiers) only 65% of the original 120 entries survived, with 135 new op-amps added.

We have used this opportunity to respond to readers' suggestions and to our own experiences using and teaching from the first edition. Thus we have rewritten the chapter on FETs (it was too complicated) and repositioned it before the chapter on op-amps (which are increasingly of FET construction). We have added a new chapter on low-power and micropower design (both analog and digital), a field both important and neglected. Most of the remaining chapters have been extensively revised. We have added many new tables, including A/D and D/A converters, digital logic components, and low-power devices, and throughout the book we have expanded the number of figures. The book now contains 78 tables (available separately as *The Horowitz and Hill Component Selection Tables*) and over 1000 figures.

Throughout the revision we have strived to retain the feeling of informality and easy access that made the first edition so successful and popular, both as reference and text. We are aware of the difficulty students often experience when approaching electronics for the first time: the field is densely interwoven, and there is no path of learning that takes you, by logical steps, from neophyte to broadly competent designer. Thus we have added extensive cross-referencing throughout the text; in addition, we have expanded the separate *Laboratory Manual* into a *Student Manual* (*Student Manual for The Art of Electronics*, by Thomas C. Hayes and Paul Horowitz), complete with additional worked examples of circuit designs, explanatory material, reading assignments, laboratory exercises, and solutions to selected problems. By offering a student supplement, we have been able to keep this volume concise and rich with detail, as requested by our many readers who use the volume primarily as a reference work.

We hope this new edition responds to all our readers' needs – both students and practicing engineers. We welcome suggestions and corrections, which should be addressed directly to Paul Horowitz, Physics Department, Harvard University, Cambridge, MA 02138.

In preparing this new edition, we are appreciative of the help we received from Mike Aronson and Brian Matthews (AOX, Inc.), John Greene (University of Cape Town), Jeremy Avigad and Tom Hayes (Harvard University), Peter Horowitz (EVI, Inc.), Don Stern, and Owen Walker. We thank Jim Mobley for his excellent copyediting, Sophia Prybylski and David Tranah of Cambridge University Press for their encouragement and professional dedication, and the never-sleeping typesetters at Rosenlaui Publishing Services, Inc. for their masterful composition in TEX.

Finally, in the spirit of modern jurisprudence, we remind you to read the legal notice here appended.

*Paul Horowitz*
*Winfield Hill*
*March 1989*

### *Legal notice*

**In this book we have attempted to teach the techniques of electronic design, using circuit examples and data that we believe to be accurate. However, the examples, data, and other information are intended solely as teaching aids and should not be used in any particular application without independent testing and verification by the person making the application. Independent testing and verification are especially important in any application in which incorrect functioning could result in personal injury or damage to property.**

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# PREFACE TO THE THIRD EDITION

Moore's Law continues to assert itself, unabated, since the publication of the second edition a quarter century ago. In this new third (and final!) edition we have responded to this upheaval with major enhancements:

- an emphasis on devices and circuits for *A/D* and *D/A conversion* (Chapter 13), because embedded microcontrollers are everywhere
- illustration of specialized peripheral ICs for use with microcontrollers (Chapter 15)
- detailed discussions of logic family choices, and of interfacing logic signals to the real world (Chapters 10 and 12)
- greatly expanded treatment of important topics in the essential analog portion of instrument design:
  - precision circuit design (Chapter 5)
  - low-noise design (Chapter 8)
  - power switching (Chapters 3, 9, and 12)
  - power conversion (Chapter 9)

And we have added many entirely new topics, including:

- digital audio and video (including cable and satellite TV)
- transmission lines
- circuit simulation with SPICE
- transimpedance amplifiers
- depletion-mode MOSFETs
- protected MOSFETs
- high-side drivers
- quartz crystal properties and oscillators
- a full exploration of JFETs
- high-voltage regulators
- optoelectronics
- power logic registers
- delta–sigma converters
- precision multislope conversion
- memory technologies
- serial buses
- illustrative "Designs by the Masters"

In this new edition we have responded, also, to the reality that previous editions have been enthusiastically embraced by the community of practicing circuit designers, even though *The Art of Electronics* (now 35 years in print) originated as a course textbook. So we've continued the "how *we* do it" approach to circuit design; and we've expanded the depth of treatment, while (we hope) retaining the easy access and explanation of basics. At the same time we have split off some of the specifically course-related teaching and lab material into a separate *Learning the Art of Electronics* volume, a substantial expansion of the previous edition's companion *Student Manual for The Art of Electronics*.[1]

Digital oscilloscopes have made it easy to capture, annotate, and combine measured waveforms, a capability we have exploited by including some 90 'scope screenshots illustrating the behavior of working circuits. Along with those doses of reality, we have included (in tables and graphs) substantial quantities of highly useful measured data – such as transistor noise and gain characteristics ($e_n$, $i_n$, $r_{bb'}$; $h_{fe}$, $g_m$, $g_{oss}$), analog switch characteristics ($R_{ON}$, $Q_{inj}$, capacitance), op-amp input and output characteristics ($e_n$ and $i_n$ over frequency, input common-mode range, output swing, auto-zero recovery, distortion, available packages), and approximate prices (!) – the sort of data often buried or omitted in datasheets but which you need (and don't have the time to measure) when designing circuits.

We've worked diligently, over the 20 years it has taken to prepare this edition, to include important circuit design information, in the form of some 350 graphs, 50 photographs, and 87 tables (listing more than 1900 active components), the last enabling intelligent choice of circuit components by listing essential characteristics (both specified and measured) of available parts.

Because of the significant expansion of topics and depth of detail, we've had to leave behind some topics that were treated in the second edition,[2] notwithstanding the use of larger pages, more compact fonts, and most figures sized to fit in a single column. Some additional related material that we had hoped to include in this volume (on real-world properties of components, and advanced topics in BJTs, FETs, op-amps, and power control) will instead be published in a forthcoming companion volume, *The Art*

[1] Both by Hayes, T. and Horowitz, P., Cambridge University Press, 1989 and 2016.

[2] Which, however, will continue to be available as an e-book.

*of Electronics: The x-Chapters*. References in this volume to those x-chapter sections and figures are set in italics. A newly updated `artofelectronics.com` website will provide a home for a continuation of the previous edition's collections of *Circuit ideas* and *Bad circuits*; it is our hope that it will become a community, also, for a lively electronic circuit forum.

As always, we welcome corrections and suggestions (and, of course, fan mail), which can be sent to horowitz@physics.harvard.edu or to hill@rowland.harvard.edu.

**With gratitude.** Where to start, in thanking our invaluable colleagues? Surely topping the list is David Tranah, our indefatigable editor at the Cambridge University Press mother-ship, our linchpin, helpful LaTeXpert, wise advisor of all things bookish, and (would you believe?) *compositor*! This guy slogged through 1,905 pages of marked-up text, retrofitting the LaTeX source files with corrections from multiple personalities, then entering a few thousand index entries, and making it all work with its 1,500+ linked figures and tables. And then putting up with a couple of fussy authors. We are totally indebted to David. We owe him a pint of ale.

We are grateful to Jim Macarthur, circuit designer extraordinaire, for his careful reading of chapter drafts, and invariably helpful suggestions for improvement; we adopted every one. Our colleague Peter Lu taught us the delights of Adobe Illustrator, and appeared at a moment's notice when we went off the rails; the book's figures are testament to the quality of his tutoring. And our always-entertaining colleague Jason Gallicchio generously contributed his master Mathematica talents to reveal graphically the properties of delta–sigma conversion, nonlinear control, filter functions; he left his mark, also, in the microcontroller chapter, contributing both wisdom and code.

For their many helpful contributions we thank Bob Adams, Mike Burns, Steve Cerwin, Jesse Colman, Michael Covington, Doug Doskocil, Jon Hagen, Tom Hayes, Phil Hobbs, Peter Horowitz, George Kontopidis, Maggie McFee, Curtis Mead, Ali Mehmed, Angel Peterchev, Jim Phillips, Marco Sartore, Andrew Speck, Jim Thompson, Jim van Zee, GuYeon Wei, John Willison, Jonathan Wolff, John Woodgate, and Woody Yang. We thank also others whom (we're sure) we've here overlooked, with apologies for the omission. Additional contributors to the book's content (circuits, inspired web-based tools, unusual measurements, etc., from the likes of Uwe Beis, Tom Bruhns, and John Larkin) are referenced throughout the book in the relevant text.

Simon Capelin has kept us out of the doldrums with his unflagging encouragement and his apparent inability to scold us for missed deadlines (our contract called for delivery of the finished manuscript in December. . . of *1994!* We're only 20 years late). In the production chain we are indebted to our project manager Peggy Rote, our copy editor Vicki Danahy, and a cast of unnamed graphic artists who converted our pencil circuit sketches into beautiful vector graphics.

We remember fondly our late colleague and friend Jim Williams for wonderful insider stories of circuit failures and circuit conquests, and for his take-no-prisoners approach to precision circuit design. His no-bullshit attitude is a model for us all.

And finally, we are forever indebted to our loving, supportive, and ever-tolerant spouses Vida and Ava, who suffered through decades of abandonment as we obsessed over every detail of our second encore.

**A note on the tools.** Tables were assembled in Microsoft Excel, and graphical data was plotted with Igor Pro; both were then beautified with Adobe Illustrator, with text and annotations in the sans-serif Helvetica Neue LT typeface. Oscilloscope screenshots are from our trusty Tektronix TDS3044 and 3054 "lunchboxes," taken to finishing school in Illustrator, by way of Photoshop. The photographs in the book were taken primarily with two cameras: a Calumet Horseman $6\times9$ cm view camera with a 105 mm Schneider Symmar $f/5.6$ lens and Kodak Plus-X 120 roll film (developed in Microdol-X 1:3 at 75°F and digitized with a Mamiya multiformat scanner), and a Canon 5D with a Scheimpflug[3]-enabling 90 mm tilt-shift lens. The authors composed the manuscript in LaTeX, using the PCTeX software from Personal TeX, Incorporated. The text is set in the Times New Roman and Helvetica typefaces, the former dating from 1931,[4] the latter designed in 1957 by Max Miedinger.

*Paul Horowitz*
*Winfield Hill*
*January 2015*
*Cambridge, Massachusetts*

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[3] What's *that*? Google it!

[4] Developed in response to a criticism of the antiquated typeface in *The Times* (London).

## Legal Notice Addendum

**In addition to the Legal Notice appended to the Preface to the Second Edition, we also make no representation regarding whether use of the examples, data, or other information in this volume might infringe others' intellectual property rights, including US and foreign patents. It is the reader's sole responsibility to ensure that he or she is not infringing any intellectual property rights, even for use which is considered to be experimental in nature. By using any of the examples, data, or other information in this volume, the reader has agreed to assume all liability for any damages arising from or relating to such use, regardless of whether such liability is based on intellectual property or any other cause of action, and regardless of whether the damages are direct, indirect, incidental, consequential, or any other type of damage. The authors and publisher disclaim any such liability.**

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